



SYSTEMS ANALYSIS AND DESIGN

IS104





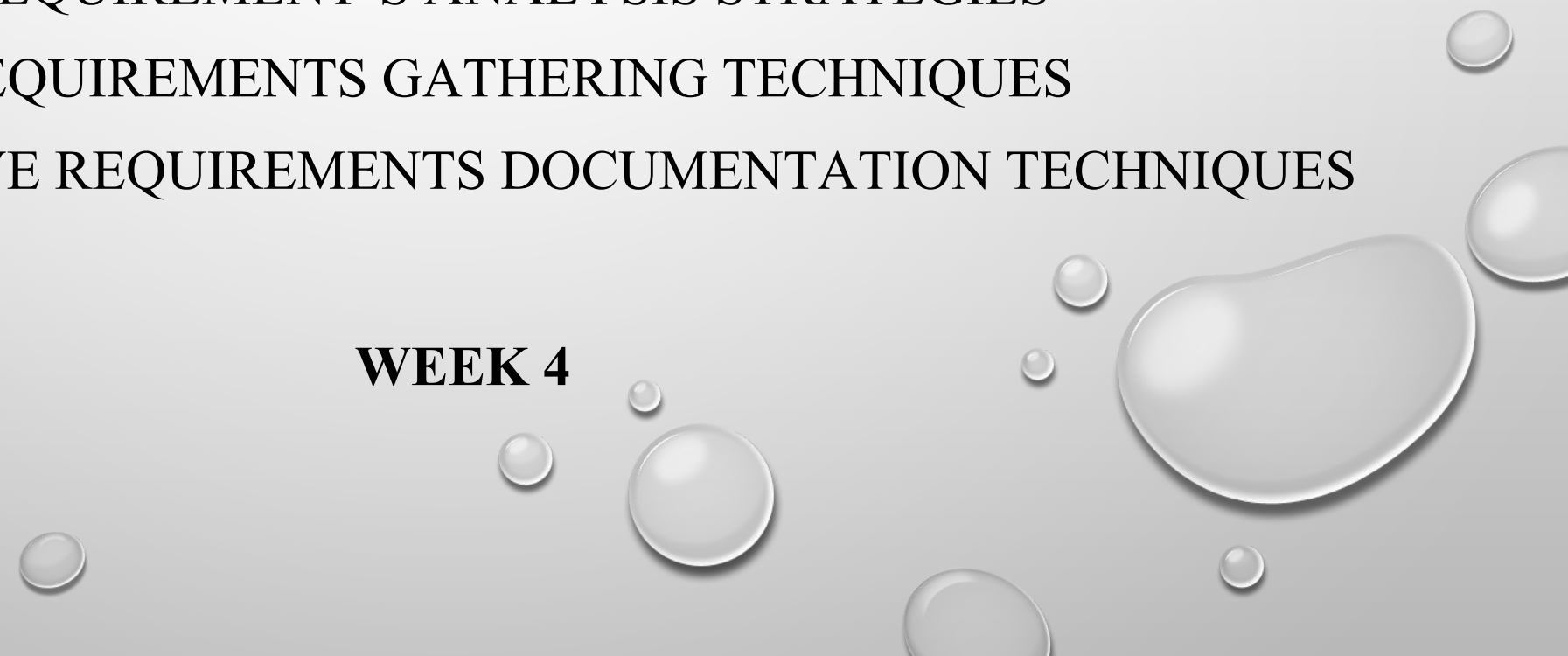
REQUIREMENTS DETERMINATION

REQUIREMENT'S ANALYSIS STRATEGIES

REQUIREMENTS GATHERING TECHNIQUES

ALTERNATIVE REQUIREMENTS DOCUMENTATION TECHNIQUES

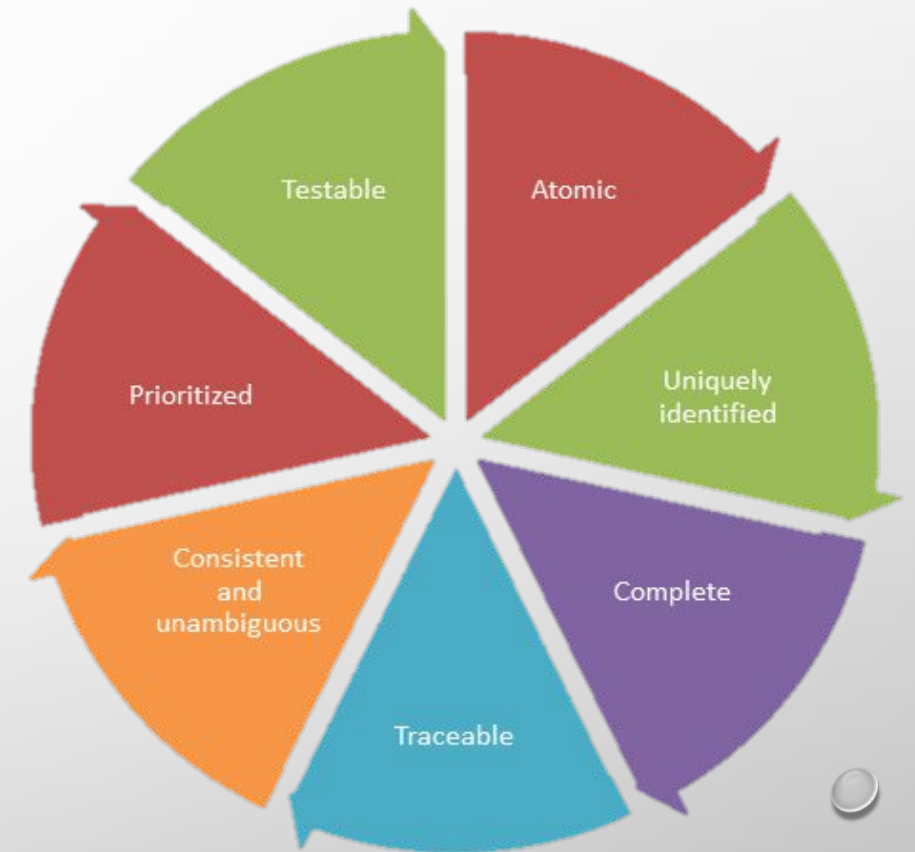
WEEK 4



Introduction

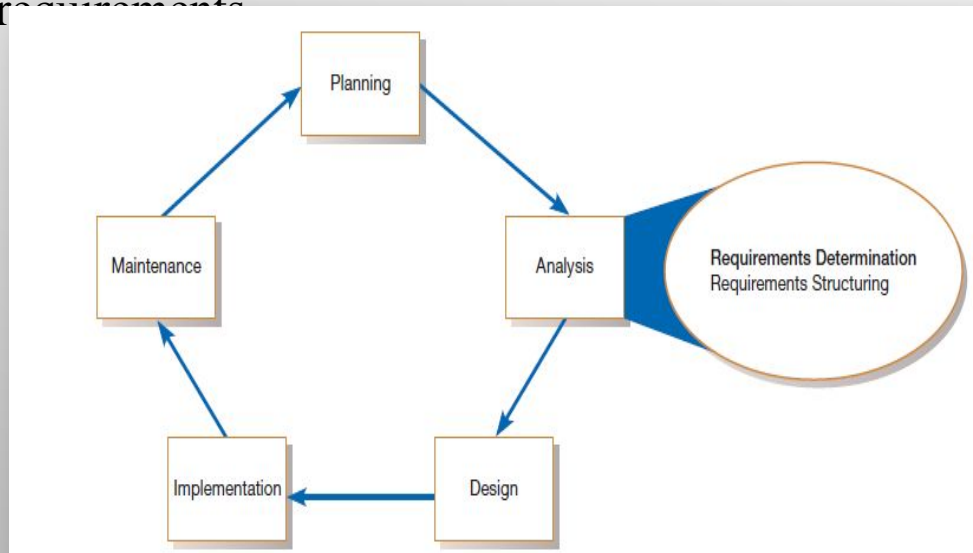
Systems analysis is a step in the systems development life cycle that involves determining how the current information system works and determining what users want in a new system. There are two parts to analysis: determining requirements and structuring requirements.

The concept of a requirement is outlined, as well as numerous types of requirements. The requirements definition statement's purpose and structure are described. Interviews, JAD sessions, questionnaires, document analysis, and observation are some of the techniques used to elicit needs. Finally, a number of requirements analysis methodologies are discussed in order to assist the analyst in identifying requirements.



PERFORMING REQUIREMENT DETERMINATION

The analysis phase is named after the term analysis, which refers to the process of breaking down a whole into its constituent pieces in order to comprehend its nature, function, and interrelationships. During the analysis phase, the systems analyst meets with the new system's business users to better understand their requirements.



The basic process of analysis involves three steps:

- Understand the existing situation (the as-is system).
- Identify improvements.
- Define requirements for the new system (the to-be system).

What is Requirement??

- ☐ A statement of what the system must do or a characteristic it must have.
- ☐ Will later evolve into a technical description of how the system will be implemented.

PERFORMING REQUIREMENT DETERMINATION

Good Systems Analyst Characteristics:

- Impertinence
- Impartiality
- Relaxing constraints
- Attention to details
- Reframing

Types of Requirements

- ☐ Functional: relates to a process or data
- ☐ Non-functional: relates to performance or usability



PERFORMING REQUIREMENT DETERMINATION

Creating a Requirements Definition

- ☐ Determine the types of functional and non-functional requirements applicable to the project.
- ☐ Use requirements-gathering techniques to collect details.
- ☐ Analysts work with users to verify, change and prioritize each requirement.
- ☐ Continue this process through analysis workflow, but be careful of scope creep.
- ☐ Requirements that meet a need but are not within the current scope can be added to a list of future enhancements.



Sample of Requirements Definition

Nonfunctional Requirements

1. Operational Requirements

- 1.1. The system will operate in Windows environment.
- 1.2. The system should be able to connect to printers wirelessly.
- 1.3. The system should automatically back up at the end of each day.

2. Performance Requirements

- 2.1. The system will store a new appointment in 2 seconds or less.
- 2.2. The system will retrieve the daily appointment schedule in 2 seconds or less.

3. Security Requirements

- 3.1. Only doctors can set their availability.
- 3.2. Only a manager can produce a schedule.

4. Cultural and Political Requirements

- 4.1. No special cultural and political requirements are anticipated.

Functional Requirements

1. Manage Appointments

- 1.1. Patient makes new appointment.
- 1.2. Patient changes appointment.
- 1.3. Patient cancels appointment.

2. Produce Schedule

- 2.1. Office Manager checks daily schedule.
- 2.2. Office Manager prints daily schedule.

3. Record Doctor Availability

- 3.1. Doctor updates schedule

PERFORMING REQUIREMENT DETERMINATION

Deliverables for Requirements Determination

From interviews and observations

- ☐ interview transcripts, observation notes, meeting minutes.

From existing written documents

- ☐ mission and strategy statements, business forms, procedure manuals, job descriptions, training manuals, system documentation, flowcharts.

From computerized sources

- ☐ reports from existing systems,
- ☐ displays and reports from system prototype



Requirements Analysis Strategies

Business Process Automation (BPA)

- ☐ Least amount of change to the current system
- ☐ Use computer technology to automate some portions

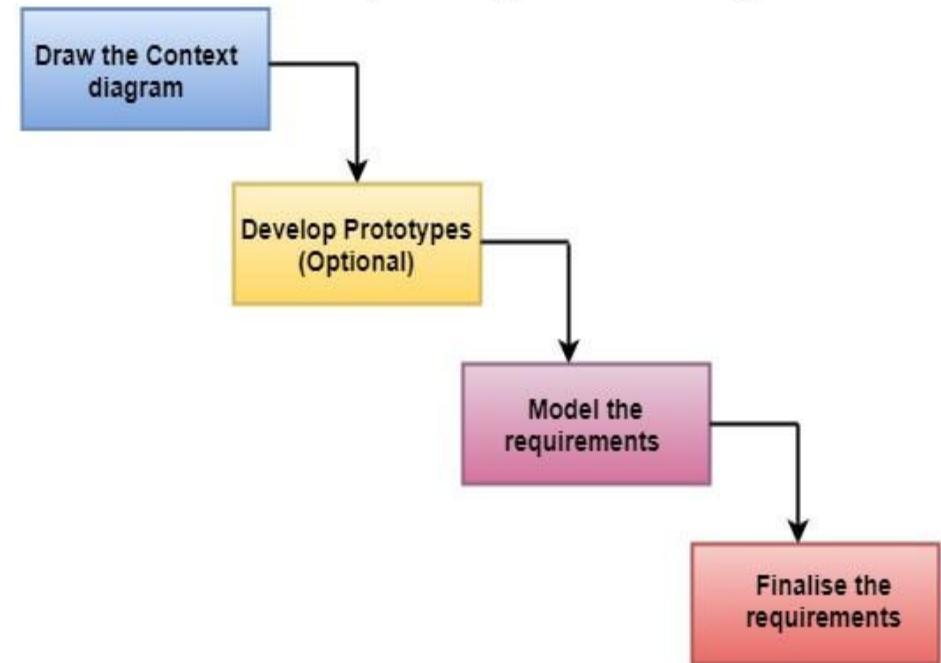
Business Process Improvement (BPI)

- ☐ Moderate amount of change is required
- ☐ Designed to improve efficiency of the current system

Business Process Reengineering (BPR)

- ☐ Most amount of change—a complete makeover
- ☐ Focus is on the to-be system—little time spent on the current system

Steps of Requirements Analysis



REQUIREMENTS ELICITATION TECHNIQUES

1. Interviews
2. JAD sessions
3. Questionnaires
4. document analysis
5. observation

1. Interview

Five basic steps to the interview process:

- ☐ selecting interviewees
- ☐ designing interview questions
- ☐ preparing for the interview
- ☐ conducting the interview
- ☐ post interview follow-up

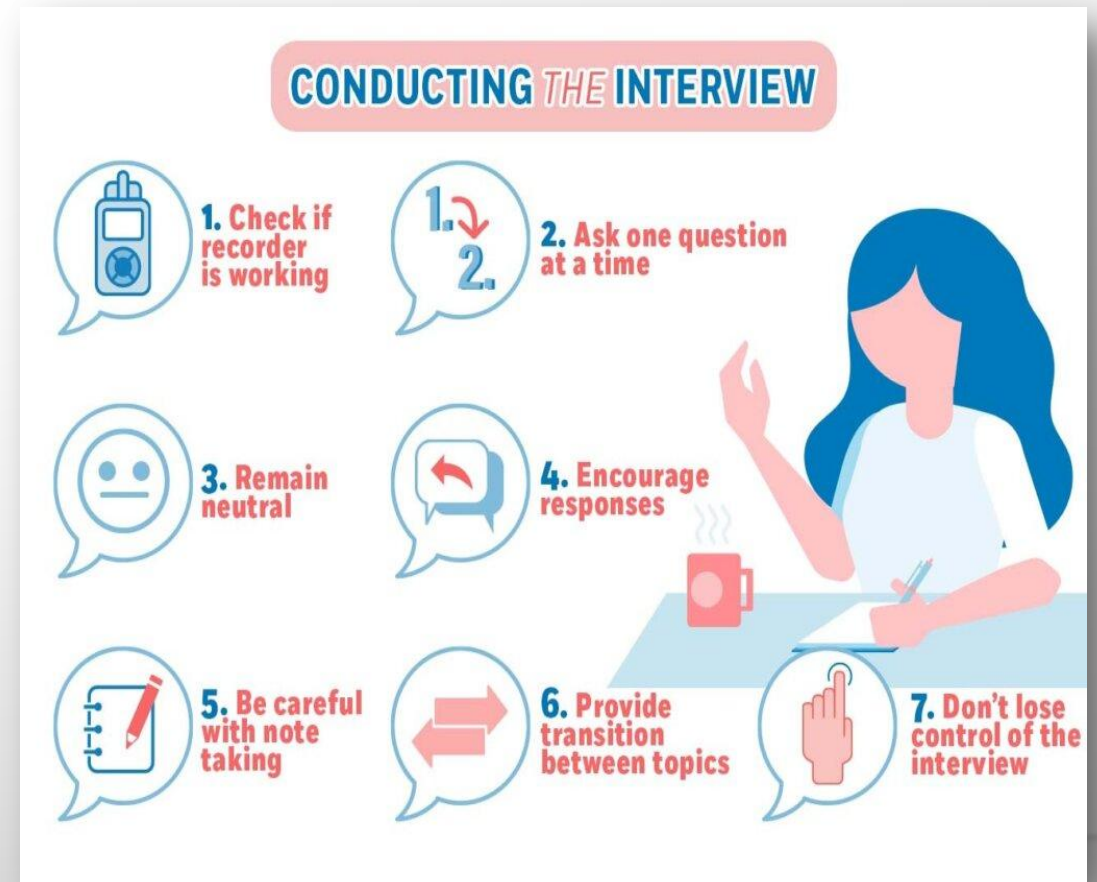


REQUIREMENTS ELICITATION TECHNIQUES

Guidelines for Effective Interviewing

Plan the interview.

- Prepare interviewee: appointment, priming questions.
 - Prepare agenda, checklist, questions.
-
- ☐ Listen carefully and take notes (tape record if permitted).
 - ☐ Review notes within 48 hours.
 - ☐ Be neutral.
 - ☐ Seek diverse views.



Typical interview guide

Interview Outline	
Interviewee: <i>Name of person being interviewed</i>	Interviewer: <i>Name of person leading interview</i>
Location/Medium: <i>Office, conference room, or phone number</i>	Appointment Date: Start Time: End Time:
Objectives: <i>What data to collect On what to gain agreement What areas to explore</i>	Reminders: <i>Background/experience of interviewee Known opinions of interviewee</i>
Agenda: Introduction Background on Project Overview of Interview Topics to Be Covered Permission to Record Topic 1 Questions Topic 2 Questions ... Summary of Major Points Questions from Interviewee Closing	Approximate Time: 1 minute 2 minutes 1 minute 5 minutes 7 minutes ... 2 minutes 5 minutes 1 minute

General Observations: <i>Interviewee seemed busy probably need to call in a few days for follow-up questions because he gave only short answers. PC was turned off-probably not a regular PC user.</i>	
Unresolved Issues, Topics Not Covered: <i>He needs to look up sales figures from 1999. He raised the issue of how to handle returned goods, but we did not have time to discuss.</i>	
Interviewee:	Date:
Questions:	Notes:
<i>When to ask question, if conditional</i> Question: 1 <i>Have you used the current sales tracking system? If so, how often?</i>	Answer <i>Yes, I ask for a report on my product line weekly.</i> Observations <i>Seemed anxious-may be overestimating usage frequency.</i> <i>If yes, go to Question 2</i>
Question: 2 <i>What do you like least about the system?</i>	Answer <i>Sales are shown in units, not dollars.</i> Observations <i>System can show sales in dollars, but user does not know this.</i>

REQUIREMENTS ELICITATION TECHNIQUES

Types of Questions

Open-ended questions

- ✓ questions that have no prespecified answers.

Closed-ended questions

- ✓ questions that ask those responding to choose from among a set of specified responses.

Probing questions

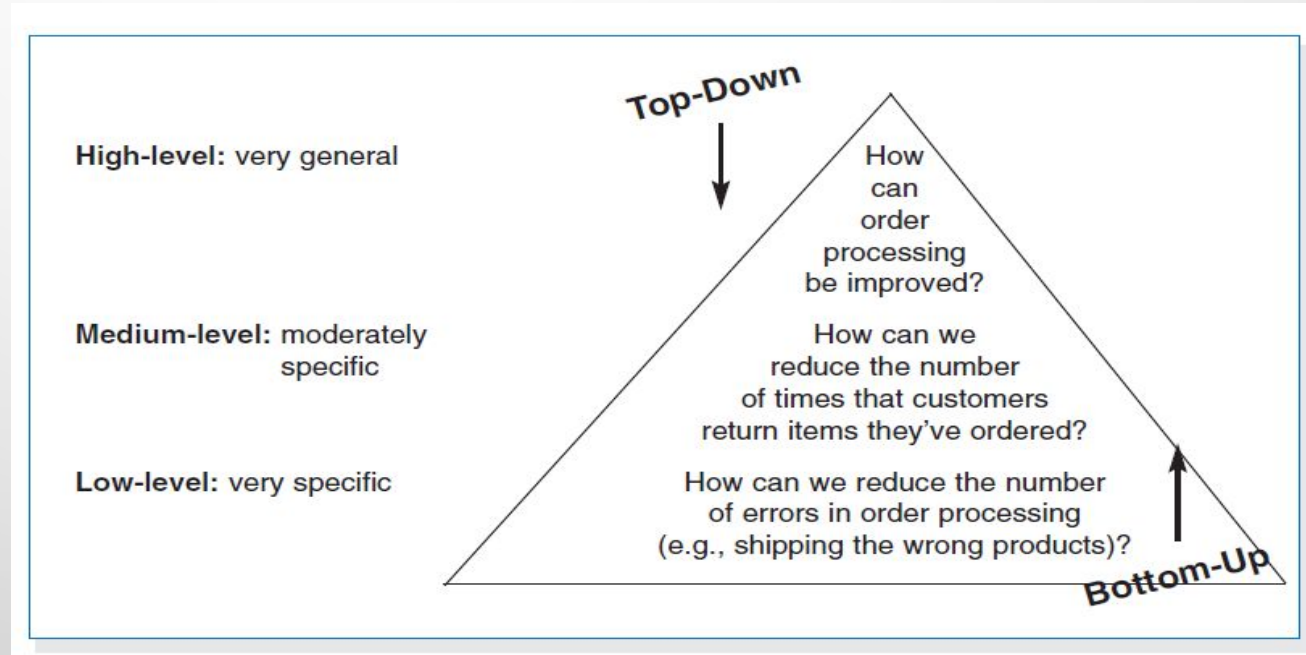
- ✓ follow up on what has just been discussed in order for the interviewer to learn more, and they often are used when the interviewer is unclear about an interviewee's answer.

Types of Questions	Examples
Closed-Ended Questions	• How many telephone orders are received per day? • How do customers place orders? • What information is missing from the monthly sales report?
Open-Ended Questions	• What do you think about the way invoices are currently processed? • What are some of the problems you face on a daily basis? • What are some of the improvements you would like to see in the way invoices are processed?
Probing Questions	• Why? • Can you give me an example? • Can you explain that in a bit more detail?

REQUIREMENTS ELICITATION TECHNIQUES

Two fundamental approaches to organizing the interview questions:

- ☐ top-down
- ☐ or bottom-up



REQUIREMENTS ELICITATION TECHNIQUES

Post-Interview

- Prepare notes and send to the interviewee for verification

Interview Notes Approved by: Linda Estey
Person Interviewed: Linda Estey, Director, Human Resources Interviewer: Barbara Wixom Purpose of Interview: • Understand reports produced for Human Resources by the current system • Determine information requirements for future system Summary of Interview: • Sample reports of all current HR reports are attached to this report. The information that is not used and missing information are noted on the reports. • Two biggest problems with the current system are: 1. The data are too old (the HR Department needs information within two days of month end; currently information is provided to them after a three-week delay) 2. The data are of poor quality (often reports must be reconciled with departmental HR database) • The most common data errors found in the current system include incorrect job level information and missing salary information. Open Items: • Get current employee roster report from Mary Skudrna (extension 4355). • Verify calculations used to determine vacation time with Mary Skudrna. • Schedule interview with Jim Wack (extension 2337) regarding the reasons for data quality problems. Detailed Notes: See attached transcript.

REQUIREMENTS ELICITATION TECHNIQUES

Benefits:

- Interactive discussion with stakeholders.
- The immediate follow-up to ensure the interviewer's understanding.
- Encourage participation and build relationships by establishing rapport with the stakeholder.

Drawbacks:

- Time is required to plan and conduct interviews.
- Commitment is required from all the participants.
- Sometimes training is required to conduct effective interviews.

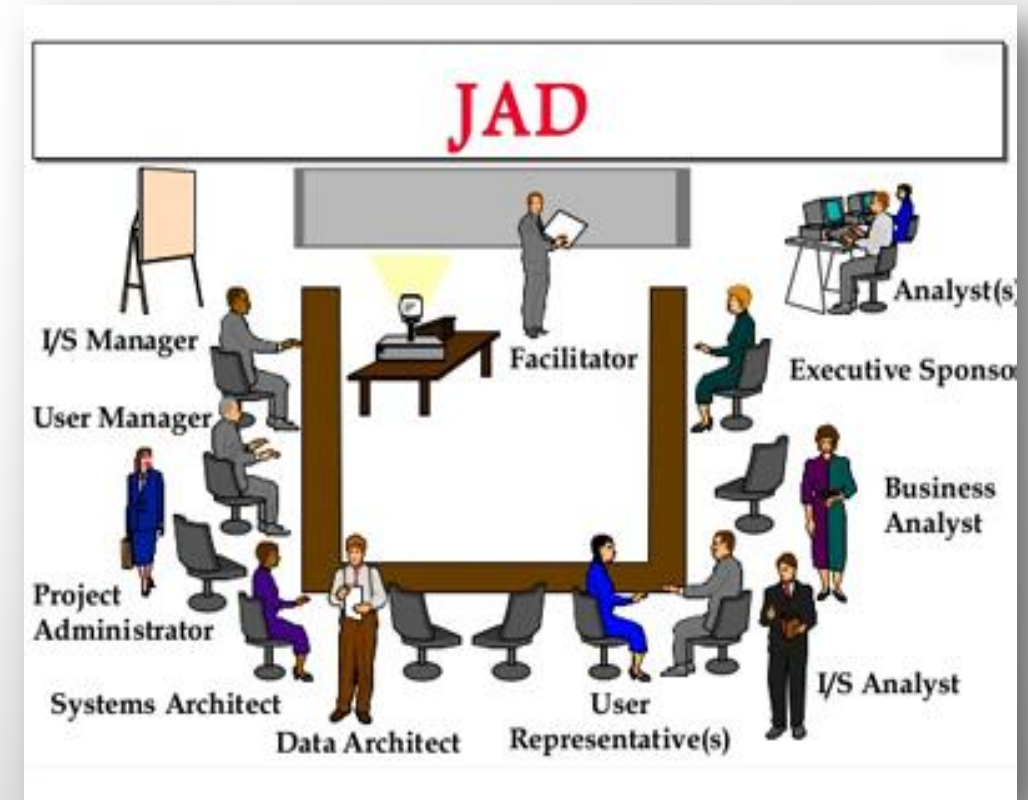


REQUIREMENTS ELICITATION TECHNIQUES

2. Joint Application Development (JAD)

Joint application development (or JAD as it is more commonly known) is an information gathering technique that allows the project team, users, and management to work together to identify requirements for the system application through a series of collaborative workshops known as

JAD sessions. It's also possible to create a prototype.



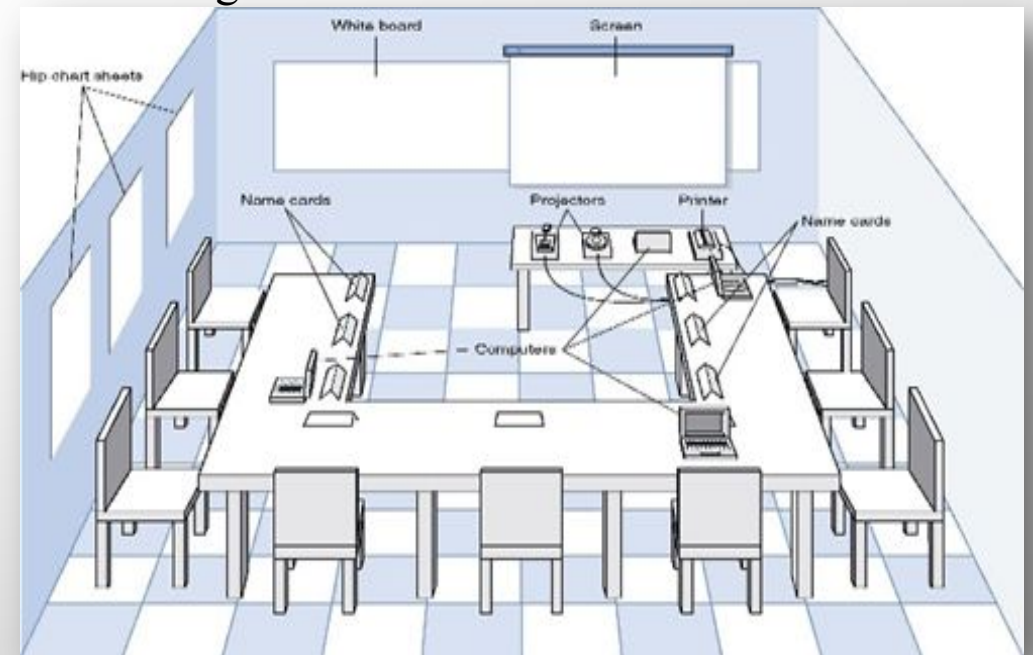
REQUIREMENTS ELICITATION TECHNIQUES

Conducting the JAD Session

Session activities may include:

- Task briefing. Provide a clear statement of what the session is to achieve.
- Problem analysis. Reach a complete understanding of the problem.
- Product description. Describe the external behavior of the target system from the user's point of view.
- Evaluation. Establish if the requirements are complete correct, consistent and implementable
- Summary. Summarize the requirements discovered

Joint Application Design (JAD) Setting



REQUIREMENTS ELICITATION TECHNIQUES

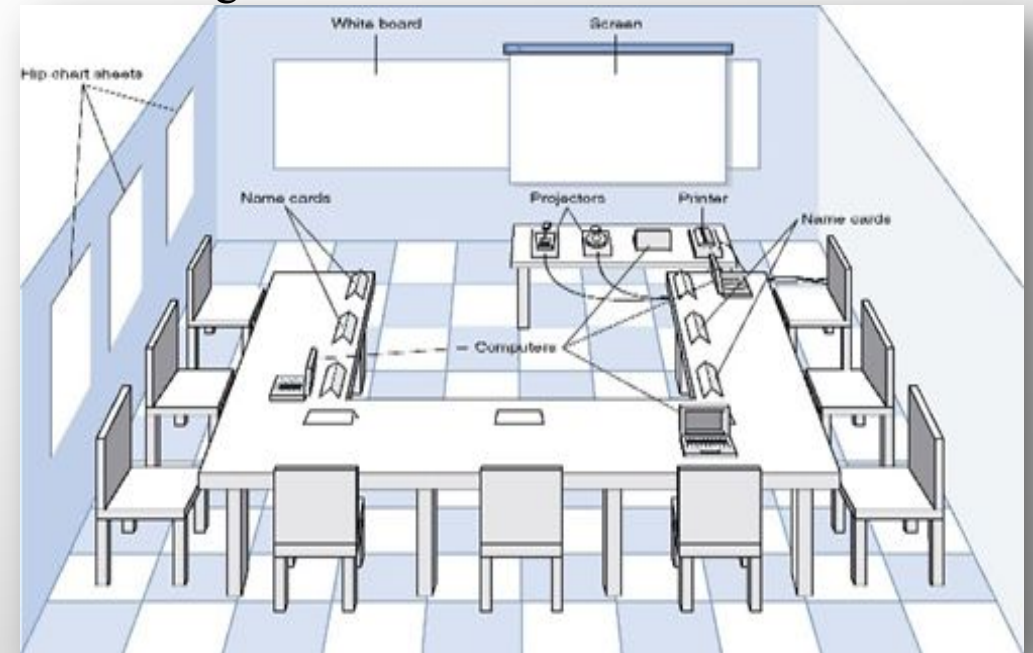
Benefits:

- Documentation is completed within hours and is provided quickly back to participants for review.
- You can get on the spot confirmation on requirements.
- Successfully gathered requirements from a large group in a short period.
- Consensus can be achieved as issues and questions are asked in the presence of all the stakeholders.

Drawbacks:

- Stakeholder's availability might ruin the session. The success rate depends on the expertise of the facilitator.
- A workshop motive cannot be achieved if there are too many participants.

Joint Application Design (JAD) Setting



REQUIREMENTS ELICITATION TECHNIQUES

3. Questionnaires

A questionnaire is a set of written questions for obtaining information from individuals. Questionnaires often are used when there is a large number of people from whom information and opinions are needed.

- ☐ A set of written questions used to obtain information from individuals
- ☐ May be paper based or electronic (e.g., web based)
- ☐ Common uses:
 - ☐ Large numbers of people
 - ☐ Need both information and opinions
 - ☐ When designing for use outside the organization (customers, vendors, etc.)
- ☐ Typical response rates: < 50% (paper); < 30% (Web)



● REQUIREMENTS ELICITATION

● TECHNIQUES

Questionnaire Steps

- ☐ Select the participants
- ☐ Identify the population
- ☐ Use representative samples for large populations
- ☐ Designing the questionnaire
- ☐ Careful question selection
- ☐ Remove ambiguities
- ☐ Administering the questionnaire
- ☐ Working to get good response rate
- ☐ Offer an incentive (e.g., a free pen)
- ☐ Questionnaire follow-up
- ☐ Send results to participants
- ☐ Send a thank-you



● REQUIREMENTS ELICITATION ● TECHNIQUES

Good Questionnaire Design

- ✓ Begin with non-threatening and interesting questions
- ✓ Group items into logically coherent sections
- ✓ Do not put important items at the very end of the questionnaire
- ✓ Do not crowd a page with too many items
- ✓ Avoid abbreviations
- ✓ Avoid biased or suggestive items or terms
- ✓ Number questions to avoid confusion
- ✓ Pretest the questionnaire to identify confusing questions
- ✓ Provide anonymity to respondents



● REQUIREMENTS ELICITATION ● TECHNIQUES

Benefits:

- Easy to get data from a large audience.
- Less time is required for the participants to respond.
- You can get more accurate information as compared to interviews.

Drawback:

- All the Stakeholders might not participate in the surveys.
- Questions may not be clear to all the participants.
- Open-ended questions require more analysis.
- Follow up surveys might be required based on the responses provided by participants.



● REQUIREMENTS ELICITATION ● TECHNIQUES

4. Document Analysis

- ☐ Provides information about the “as-is” system
- ☐ Review technical documents when available
- ☐ Review typical user documents:
 - ✓ Forms
 - ✓ Reports
- ☐ Policy manuals
- ☐ Look for user additions to forms
- ☐ Look for unused form elements



● REQUIREMENTS ELICITATION ● TECHNIQUES

Benefits:

- Existing documents can be used to compare current and future processes.
- Existing documents can be used as a base for future analysis.

Drawbacks:

- Existing documents might not be updated.
- Existing documents might be completely outdated.
- Resources worked on the existing documents might not be available to provide information.
- This process is time-consuming.



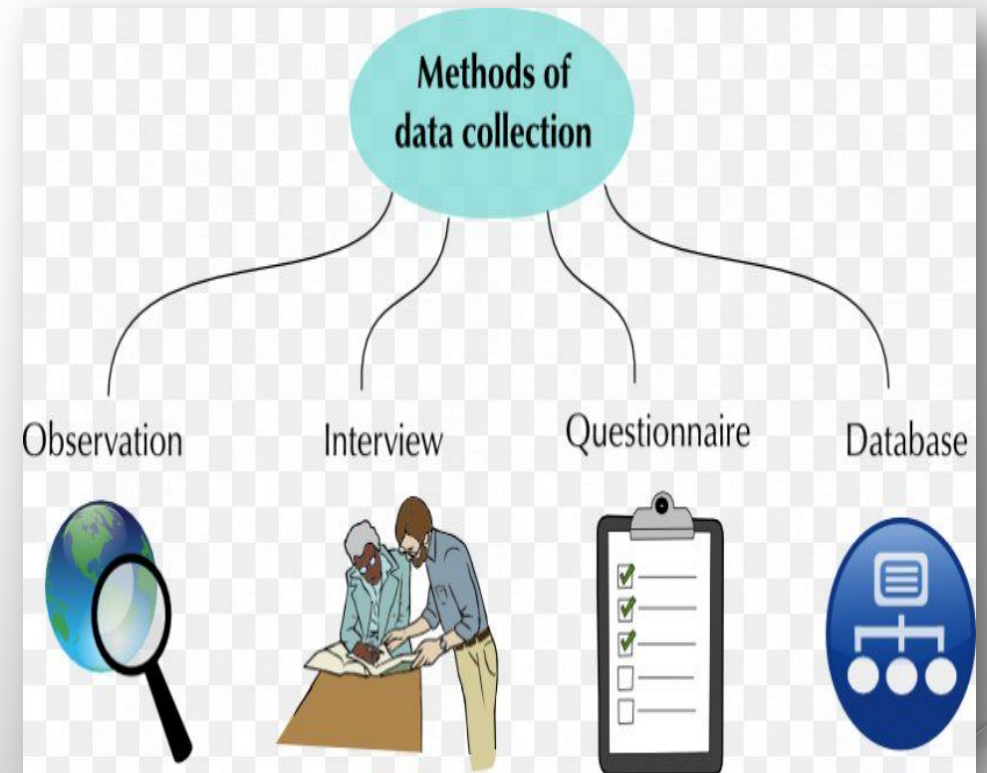
REQUIREMENTS TECHNIQUES

ELICITATION

5. Observation

The major goal of the observation session is to learn about other people's activities, tasks, tools, and events. All stakeholders are aware of the goal of the observation session, they agree on the expected outcomes, and the session meets their expectations, according to the observation plan. You must make it clear to the participants that their performance will not be evaluated.

- ✓ Users/managers often don't remember everything they do
- ✓ Checks validity of information gathered in other ways
- ✓ Behaviors may change when people are watched
- ✓ Workers tend to be very careful when watched
- ✓ Keep a low profile
- ✓ Try not to interrupt or influence workers
- ✓ Be careful not to ignore periodic activities
- ✓ Weekly ... Monthly ... Annually



REQUIREMENTS TECHNIQUES

ELICITATION

Two types of observation

Active observation

- ✓ to ask questions and try to attempt the work that other persons are doing.

Passive observation

- ✓ silent observation i.e., you sit with others and just observe how they are doing their work without interpreting them.

Benefits:

- ✓ The observer will get a practical insight into the work.
- ✓ Improvement areas can be easily identified.

Drawbacks:

- ✓ Participants might get disturbed.
- ✓ Participants might change their way of working during observation and the observer might not get a clear picture.
- ✓ Knowledge-based activities cannot be observed.

Requirements-Gathering Techniques Compared

- ☐ A combination of techniques may be used
- ☐ Document analysis & observation require little training; JAD sessions can be very challenging

	Interviews	Joint Application Design	Questionnaires	Document Analysis	Observation
Type of information	As-is, improvements, to-be	As-is, improvements, to-be	As-is, improvements	As-is	As-is
Depth of information	High	High	Medium	Low	Low
Breadth of information	Low	Medium	High	High	Low
Integration of information	Low	High	Low	Low	Low
User involvement	Medium	High	Low	Low	Low
Cost	Medium	Low-Medium	Low	Low	Low-Medium

Alternative Techniques

- Concept Maps
 - ✓ Represent meaningful relationships between concepts
 - ✓ Focus individuals on a small number of key ideas

- Story Cards & Task Lists
 - ✓ Associated with agile development methods
 - ✓ File cards with a single requirement
 - ✓ Each requirement is discussed
 - How much effort is required to implement it
 - A task list is created for each requirement (story)
 - Large requirements can be split into smaller sections

The System Proposal

- Combines all material created in planning & analysis
- Included sections:
 - ✓ Executive summary
 - Provides all critical information in summary form
 - Helps busy executives determine which sections they need to read in more detail
 - ✓ The system requests
 - ✓ The workplan
 - ✓ The feasibility analysis
 - ✓ The requirements definition
 - ✓ Current models of the system (expected to evolve)

System Proposal Template

1. Table of Contents

2. Executive Summary

A summary of all the essential information in the proposal so a busy executive can read it quickly and decide what parts of the proposal to read in more depth.

3. System Request

The revised system request form (see Chapter 2).

4. Workplan

The original workplan, revised after having completed analysis (see Chapter 2).

5. Feasibility Analysis

A revised feasibility analysis, using the information from analysis (see Chapter 2).

6. Requirements Definition

A list of the functional and nonfunctional business requirements for the system (this chapter).

7. Functional Model

An activity diagram, a set of use case descriptions, and a use case diagram that illustrate the basic processes or external functionality that the system needs to support (see Chapter 4).

8. Structural Models

A set of CRC cards, class diagram and object diagrams that describe the structural aspects of the to-be system (see Chapter 5). This may also include structural models of the current as-is system that will be replaced.

9. Behavioral Models

A set of sequence diagrams, communication diagrams, behavioral state machines, and a CRUD matrix that describe the internal behavior of the to-be system (see Chapter 6). This may include behavioral models of the as-is system that will be replaced.

Appendices

These contain additional material relevant to the proposal, often used to support the recommended system. This might include results of a questionnaire survey or interviews, industry reports and statistics, and so on.



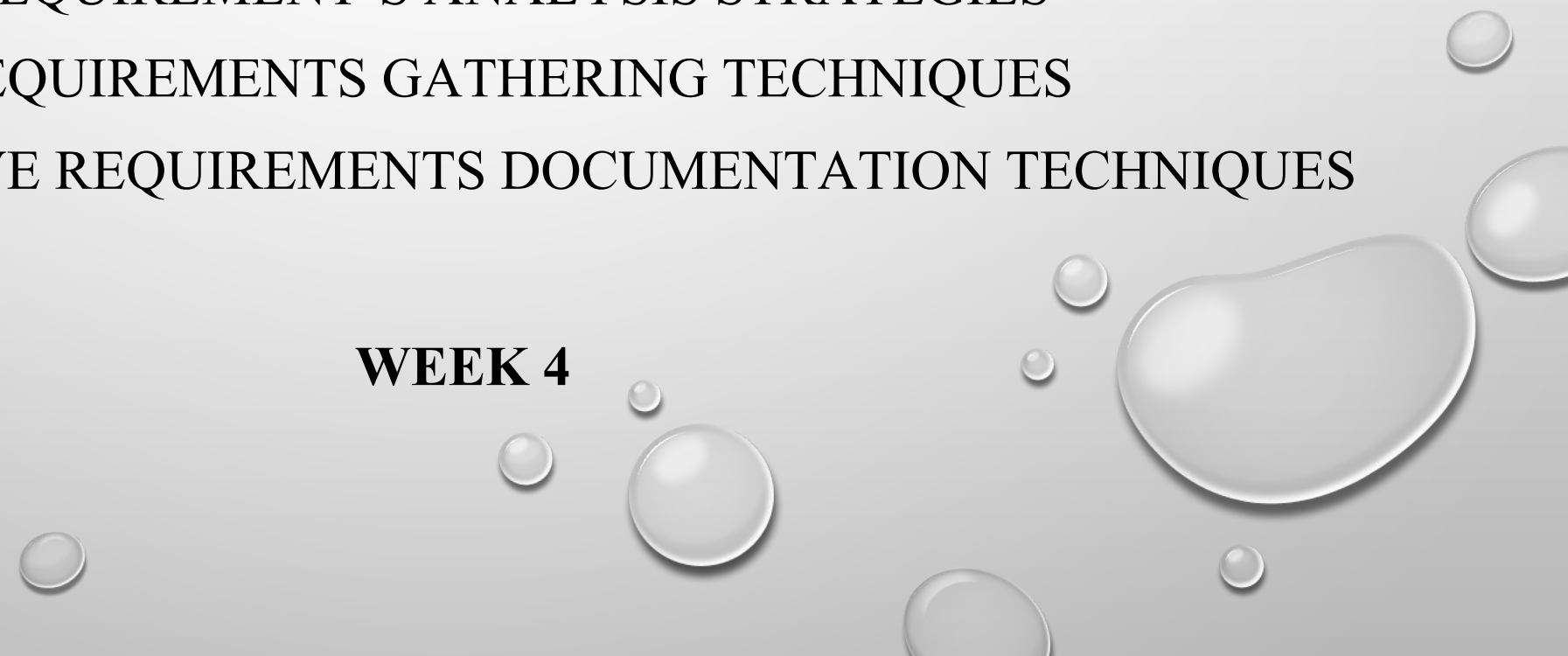
REQUIREMENTS DETERMINATION

REQUIREMENT'S ANALYSIS STRATEGIES

REQUIREMENTS GATHERING TECHNIQUES

ALTERNATIVE REQUIREMENTS DOCUMENTATION TECHNIQUES

WEEK 4



The background features a light gray gradient. In the top-left corner, there are several water droplets of varying sizes, some with highlights and shadows, giving them a 3D appearance. Similarly, in the bottom-right corner, there are more water droplets, also with realistic shading and highlights.

**THANK
YOU!**